Yixin Chen 陈亦新

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Research code: GitHub / YixinChen-AI
Publications: Google Scholar



Research profile

I work on universal medical image segmentation and cross-modality adaptation. I also study multi-organ metabolism using whole-body PET/CT.

14 first/co-first-author papers in Nature Communications, IEEE TMI, IEEE TRPMS, MICCAI and other venues (3 challenge workshop papers; 1 accepted), plus **7 other collaborative papers**. **12 patent records** (5 granted) and **1 sole-authored book**.

Education

Peking University, PhD in Medical Imaging Technology, expected 2027 — September 2023–present
Institute of Medical Technology, Health Science Center; Supervisor: Prof. Zhaoheng Xie

King's College London, MSc in Artificial Intelligence, with Distinction — 2019–2020

Xiamen University, BSc in Automation; second bachelor's degree in Mathematical Finance — 2015–2019

Research funding

NSFC Young Student Basic Research Project (Doctoral Student) — January 2026–December 2027
Principal investigator, grant 625B2004, RMB 300,000. Project: Multi-Organ Metabolic Information Extraction and Analysis Using Total-Body PET/CT.

Selected research

Universal segmentation and cross-modality adaptation

Nature Communications, 2025; Co-first author [Paper] [Code]

MPUM: Modality-projection universal model for comprehensive full-body medical imaging segmentation
Modality projection generates modality-specific kernels to segment PET, CT and MRI.

IEEE TMI, 2024; First author [Paper] [Code]

PCNet: Prior Category Network for CT Universal Segmentation Model
Anatomical category prompts and a hierarchical loss guide universal CT segmentation.

MICCAI, 2024; Poster; First author [Paper] [Code]

LUCIDA: Low-Dose Universal-Tissue CT Image Domain Adaptation for Medical Segmentation
It uses segmentation reconstruction and Fourier adaptation to segment low-dose CT tissues without target labels.

Whole-body PET/CT and multi-organ metabolism

EJNMMI, 2026; Accepted; First author

Whole-Body [^{18}F]FDG PET/CT Characterizes Multi-Organ Metabolic Alterations in MASLD
In **294** adults at two centres, **16** imaging features differed between metabolic dysfunction-associated steatotic liver disease (MASLD) and controls, with replication at the second centre.

Publications

* Equal contribution.

First- and co-first-author papers (11)

1. **Yixin Chen**, Chuanzhi Zhou, Lijuan Guo, et al. Whole-Body [^{18}F]FDG PET/CT Characterizes Multi-Organ Metabolic Alterations in MASLD. **EJNMMI***, 2026;* Accepted*.*
2. **Yixin Chen**, Yan Wang, Wenrui Shao, Zhaoheng Xie. PHATE-Net: Differentiable Pseudotime Learning for Trustworthy Disease Trajectories in PET. **CVPR Findings***, 2026, pp. 2534-2543.*
3. **Yixin Chen**[*], Lin Gao[*], Yajuan Gao, Rui Wang, Jingge Lian, Xiangxi Meng, Yanhua Duan, Leiying Chai, Hongbin Han, Zhaoping Cheng, Zhaoheng Xie. Modality-projection universal model for comprehensive full-body medical imaging segmentation. **Nature Communications***, 2025, 16:9423.* [Code]
4. **Yixin Chen**, Yan Wang, Zhaoheng Xie. VP-SFDA: Visual Prompt Source-Free Domain Adaptation for Cross-Modal Medical Image. **Health Data Science***, 2025, 5:0143.*
5. **Yixin Chen**, Yajuan Gao, Lei Zhu, Wenrui Shao, Yanye Lu, Hongbin Han, Zhaoheng Xie. PCNet: Prior Category Network for CT Universal Segmentation Model. **IEEE TMI***, 2024, 43(9):3319-3330.* [Code]
6. **Yixin Chen**, Xiangxi Meng, Yan Wang, Shuang Zeng, Xi Liu, Zhaoheng Xie. LUCIDA: Low-Dose Universal-Tissue CT Image Domain Adaptation for Medical Segmentation. **MICCAI***, 2024, LNCS 15008:393-402;* Poster*.* [Code]
7. **Yixin Chen**[*], Yajun Gao[*], Lei Zhu, Jianan Li, Yan Wang, Jiakui Hu, Hongbin Han, Yanye Lu, Zhaoheng Xie. Structure-Enhanced Unsupervised Domain Adaptation for CT Whole-Brain Segmentation. **IEEE TRPMS***, 2024, 8(8):926-938.* [Code]
8. Yan Wang[*], **Yixin Chen**[*], Yingying Zhang, Haogang Zhu. Rethinking Disentanglement in Unsupervised Domain Adaptation for Medical Image Segmentation. **IEEE EMBC***, 2023, pp. 1-6.*
9. Xixiang Lin[*], Feifei Yang[*], **Yixin Chen**[*], et al. Echocardiography-based AI for detection and quantification of atrial septal defect. **Frontiers in Cardiovascular Medicine***, 2023, 10:985657.*
10. Yan Wang[*], **Yixin Chen**[*], Wenjun Wang, Haogang Zhu. MSGAN: Multi-Stage Generative Adversarial Networks for Cross-Modality Domain Adaptation. **IEEE EMBC***, 2022, pp. 520-524.*
11. Jianan Li[*], **Yixin Chen**[*], Minyu Zhang, et al. A deep learning model system for diagnosis and management of adnexal masses. **Cancers***, 2022, 14(21):5291.*

Other collaborative papers (7)

12. Haowei Xu, **Yixin Chen**, Wanyi Fu, Hongbin Han, Zhaoheng Xie. NeuronCtrl: Geometry-Aware Safe Closed-Loop Generative Control for Neuronal Microenvironment Dynamics. **ICML***, 2026, PMLR 306:140675-140728;* Spotlight*.*
13. Wenrui Shao, **Yixin Chen**, Ourui Fu, Tian Li, Zhi Yang, Nan Li, Hongbin Han, Xiangxi Meng, Zhaoheng Xie. Neural Network Approximation Aided Kinetic Parameter Inference of Compartment Models in Dynamic PET. **IEEE TRPMS***, 2026.*
14. Wenrui Shao, Yarong Zhang, Fen Du, Fangxiao Cheng, **Yixin Chen**, Xiangxi Meng, Ying Liang, Zhaoheng Xie. Neural network-aided unsupervised input function estimation for dual-time-window PET Patlak analysis. **EJNMMI Physics***, 2025, 12(1):100.*
15. W. Shao, **Yixin Chen**, Z. Wang, R. Xu, X. Meng, Z. Xie. A Physiology-Supervised Neural Operator Framework for Kinetic Parameter Estimation in Dynamic PET. **IEEE NSS/MIC/RTSD***, 2025, p. 1.*
16. W. Shao, Y. Zhang, **Yixin Chen**, X. Meng, Y. Liang, Z. Xie. Neural Network-Asissted Dual-time-window Acquisition: Modeling Incomplete Input Functions for Patlak Analysis. **IEEE NSS/MIC/RTSD***, 2024, p. 1.*
17. Yan Wang, Jian Cheng, **Yixin Chen**, Shuai Shao, Lanyun Zhu, Zhenzhou Wu, Tao Liu, Haogang Zhu. FVP: Fourier Visual Prompting for Source-Free Unsupervised Domain Adaptation of Medical Image Segmentation. **IEEE TMI***, 2023, 42(12):3738-3751.*
18. Xixiang Lin, Feifei Yang, **Yixin Chen**, et al. Echocardiography-based AI detection of regional wall motion abnormalities and quantification of cardiac function in myocardial infarction. **Frontiers in Cardiovascular Medicine***, 2022, 9:903660.*

MICCAI challenge workshop papers (first author) (3)

19. **Yixin Chen**. Diagnosis-Conditioned Retrieval for Structured Pathology Workflows. **MICCAI REG² challenge workshop***, 2026, LNCS 17252;* Poster*.*
20. **Yixin Chen**, Yizhen Peng. Acquisition-Structured nnU-Net Ensembles for Multi-Sequence Cardiac MRI Segmentation and Quantification. **MICCAI CMRSeg workshop***, 2026, LNCS 17268;* Selected for oral presentation*.*
21. **Yixin Chen**. Failure-Mode-Driven Multimodal Mitral Valve Segmentation for the MVAA 2026 Challenge. **MICCAI MVAA challenge workshop***, 2026, LNCS 17268;* Poster*.*

Teaching and authorship

Teaching assistant, Peking University

Principles of Medical Imaging Equipment

Computer vision course instructor

Independently recorded introductory and intermediate courses in computer vision, covering machine learning fundamentals and computer vision algorithms.

Deep Learning Theory and Practice Using PyTorch 2021

深度学习理论与实战——PyTorch案例详解

Sole author. Tsinghua University Press, 2021. In Chinese. ISBN 978-7-302-56850-6.

Professional experience

AirDoc July–September 2022

Algorithm Engineer

Developed a model to predict postoperative ICL vault from ocular measurements and lens size. Worked on ultra-widefield fundus image analysis and algorithm requirements for medical device registration.

BioMind November 2020–June 2022

Medical Imaging Algorithm Engineer

Developed cardiac ultrasound models for image quality assessment, Doppler spectrum segmentation and measurement, septal defect detection and valve regurgitation analysis.

Worked with clinicians on system integration and contributed to an ultrasound system for ovarian mass assessment.

Awards and presentations

MICCAI challenge awards

autoPET V: second place in Category 1 2026

CMR-MULTI: second place 2026

REG²: fourth place 2026

CHIMERA-agent: Innovative Agent Solution Award 2026

Oral presentations

MICCAI 2026 CHIMERA-Agent Workshop 2026-10-01

MICCAI 2026 CMRSeg Workshop 2026-10-01

MICCAI 2026 autoPET V Workshop 2026-09-27

Academic service

Nature Communications, Early Career Co-Reviewer

Peking University news coverage

Doctoral student wins two awards in the MICCAI 2026 international challenges 2026-10-10

Yixin Chen awarded a 2025 NSFC doctoral research grant 2026-01-13

Patents

12 public patent records: 5 granted Chinese invention patents, 6 published Chinese invention applications, and 1 published PCT international application.

Granted Chinese invention patents

1. **Brain-region segmentation method and system for CT brain images**
   Yixin Chen (陈亦新), 谢肇恒
   Application 202310997955.8 (2023-08-09); Publication CN117649519A (2024-03-05)
   Grant CN117649519B (2026-09-11).

2. **Method and apparatus, device, and medium for generating two-dimensional cardiac ultrasound view images**
   蒲海涛, 陈晓天, 张培芳, **Yixin Chen (陈亦新)**, 叶菁, 高雨霏, 吴振洲
   Application 202211350737.7 (2022-10-31); Publication CN115511703A (2022-12-23)
   Grant CN115511703B (2023-05-09).

3. **Model-training method and apparatus, electronic device, and storage medium**
   Yixin Chen (陈亦新), 王焱, 陈晓天, 任艺柯, 张培芳, 吴振洲
   Application 202210226361.2 (2022-03-09); Publication CN115099293A (2022-09-23)
   Grant CN115099293B (2023-04-18).

4. **Method and apparatus for removing artifacts from cardiac ultrasound images**
   蒲海涛, 张培芳, 陈晓天, **Yixin Chen (陈亦新)**, 叶菁, 吴振洲
   Application 202210803955.5 (2022-07-07); Publication CN115170424A (2022-10-11)
   Grant CN115170424B (2023-04-07).

5. **Method and apparatus, electronic device, and storage medium for determining the length of an atrial septal defect**
   Yixin Chen (陈亦新), 蒲海涛, 林锡祥, 陈晓天, 张培芳, 吴振洲
   Application 202210203636.0 (2022-03-03); Publication CN114565622A (2022-05-31)
   Grant CN114565622B (2023-04-07).

Published Chinese invention applications

6. **Method and system for analyzing metabolism in an organism**
   Yixin Chen (陈亦新), 谢肇恒, 徐若妍
   Application 202510554692.2 (2025-04-29); Publication CN120458609A (2025-08-12).

7. **Neural-network-based method for biomedical parametric imaging and pharmacokinetic analysis**
   邵闻睿, 谢肇恒, 孟祥溪, **Yixin Chen (陈亦新)**
   Application 202411849298.3 (2024-12-16); Publication CN119811674A (2025-04-11).

8. **Method for analyzing differences in fundus images and related products**
   Yixin Chen (陈亦新), 赵昕
   Application 202310922867.1 (2023-07-25); Publication CN119379577A (2025-01-28).

9. **Method for processing ultra-widefield fundus images and related products**
   Yixin Chen (陈亦新), 赵昕
   Application 202310922430.8 (2023-07-25); Publication CN119379707A (2025-01-28).

10. **Method for training an image-classification model and related products**
    Yixin Chen (陈亦新), 赵昕
    Application 202310332446.3 (2023-03-30); Publication CN118736653A (2024-10-01).

11. **Data-processing system and method**
    焦文沛, 谢肇恒, **Yixin Chen (陈亦新)**
    Application 202410191474.2 (2024-02-21); Publication CN118039082A (2024-05-14).

Published PCT international application

12. **Image-processing method and apparatus, electronic device, and storage medium**
    齐秀梅, **Yixin Chen (陈亦新)**, 赵昕
    Application PCT/CN2023/106122 (2023-07-06); Publication WO2024216763A1 (2024-10-24).